CLEANUPCREW

1. Project Overview

CleanupCrew is a community-powered platform that organizes real-world cleanliness drives by connecting:

- Volunteers who want to work
- Donors who want to contribute
- Organizers who manage execution

with full transparency, tracking, and proof of impact.

Core Philosophy

“From random goodwill → to structured social action.”

The platform ensures that:

- Problems are properly reported
- Events are professionally organized
- Money usage is visible
- Volunteer work is verified
- Impact is measurable

2. Real Problem We Are Solving

Ground Reality

Cities suffer from:

- Illegal garbage dumping
- Plastic waste zones
- Dirty public spaces
- Lack of organized cleanup

But citizens face:

1. No centralized place to join drives
2. No trust in donations
3. No proof of real work
4. No organized volunteer management

CleanupCrew Becomes:

A digital bridge between:

Problem → People → Money → Action → Proof

3. User Roles & Permissions

A. Public User (Without Login)

Can:

- View all drives
- Explore map
- See reports
- View leaderboards
- Read impact stats

B. Registered User

Can:

- Become volunteer
- Donate
- Track activity
- Access dashboard
- Slot booking
- Role selection
- QR attendance
- History tracking
- Rewards & badges
- Donate to specific drive
- Track expense usage (if donated)
- Download impact report
- Leaderboard ranking

. Admin

Super control:

- Create drives
- Verify reports
- Manage attendance
- Upload expenses
- Moderate map
- User control

4. Complete System Modules

MODULE 1 – Smart Spot Reporting

User submits:

- Photo
- Location
- Description
- Severity

Improvements Added

✓ Duplicate location detection

✓ Spam prevention

✓ Admin verification

✓ Status lifecycle:

- Reported
- Verified
- Drive Created
- Cleaned

MODULE 2 – Drive Management

Admin creates drive:

- Title

- Location
- Date/time
- Max volunteers
- Funding goal
- Required roles

Role Types

- Cleaner
- Coordinator
- Photographer
- Logistics Helper

MODULE 3 – Slot Booking System

Logic

- No overbooking
- One role per user
- Cancellation before deadline
- Waitlist if full

New Addition

✓ QR based attendance

✓ Timestamp record

✓ Admin confirmation

MODULE 4 – Donation System

Smart Funding

- Drive-specific donations
- Live progress bar
- Transaction history

New Additions

✓ Expense categories

✓ Bill/proof upload

✓ Money usage breakup:

- Equipment
- Transport
- Refreshments
- Misc

MODULE 5 – Impact Measurement

After event admin enters:

- Waste collected (kg)
- Area cleaned
- Volunteer count
- Work hours

Uploads:

- Before photos
- After photos

➞ Quantifiable + visual proof

MODULE 6 – Map System

Interactive map shows:

- Reported spots
- Active drives
- Cleaned locations

Improvements

✓ Duplicate merge

✓ Status color coding

✓ History timeline

MODULE 7 – Security Layer

Added:

- Email verification
- JWT auth
- Role access control
- Duplicate accounts block
- Admin approval flow

MODULE 8 – Transparency Portal

Public can see:

- Money collected
- Expenses
- Photos
- Attendance

➞ Trust = core pillar

MODULE 9 – Gamification

Leaderboards

- Top Donors
- Top Volunteers

Rewards

- Badges
- Certificates
- Progress goals

5. Dashboards

User Dashboard

- Joined drives
- Donations
- Impact summary
- QR attendance
- Upcoming events
- Reward level
- Money usage
- Reports
- Ranking

Admin Dashboard

- Drive CRUD
- Attendance verify
- Expense upload
- Map moderation
- Analytics

6. Technology Stack

Frontend

- React
- Tailwind
- Framer Motion
- Leaflet Maps

Backend

- Node + Express
- JWT

- REST APIs

Database

- MongoDB

Payment

- Stripe

7. Database Design

Users

- id
- role
- email
- profile
- stats

Drives

- location
- slots
- funding
- status

Donations

- user
- amount
- drive
- receipt

Attendance

- QR
- time
- status

Reports

- before/after
- metrics
- expenses

8. Complete User Journey

1. User reports dirty spot
2. Admin verifies
3. Drive created
4. Volunteers book slots
5. Donors fund
6. Event day → QR attendance
7. Cleanup happens
8. Admin uploads:
- photos
- expenses
- metrics
9. Leaderboards update
10. Impact visible

➞ End-to-end transparency loop

9. Business Logic Rules

- No duplicate spots
- Max volunteer limit
- Verified expenses only
- QR mandatory
- Role-based access

10. Why This Is Strong Project

- Real world use case
- Payments

- Maps
- Security
- Analytics
- Multi-role
- Proof system

11. Viva Ready Explanation

“CleanupCrew is a transparency-focused platform that organizes locality-based cleanup drives by connecting volunteers and donors. It provides slot booking with QR attendance, drive-specific funding with expense tracking, map-based reporting, and measurable impact metrics to convert community goodwill into structured real-world cleanliness action.”